Status	Finished
Started	Wednesday, 21 October 2026, 7:27 AM
Completed	Wednesday, 21 October 2026, 7:50 AM
Duration	22 mins 40 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Jump search
b. Binary search
c. Linear search ✓
d. Hashing

Question **2**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The search is considered unsuccessful ✓
c. The first element is returned
d. The last element is returned

Question **3**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection

b. Complexity

c. Bubble ✓

d. Insertion

Question **4**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist

b. At the beginning of the list

c. In the right sublist ✓

d. In the middle of the list

Question **5**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Linear search ✓

b. Both (1) & (3)

c. Binary search

d. Hash search

Question **6**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By comparing each element in the list sequentially ✓
- b. By sorting the list first
- c. By dividing the list into halves
- d. By using a hash function

Question **7**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Linear search ✓
- b. Interpolation search
- c. Binary search
- d. Hash search

Question **8**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Binary search ✓
- b. Linear search
- c. Hash search
- d. Sequential search

Question **9**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n/2
- b. log n
- c. n-1
- d. n ✓

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The search continues in the left or right sublist ✓
c. The search continues from the beginning
d. The list is sorted

Question **11**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)
b. O(1) ✓
c. O👎
d. O(n log n)

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist
b. Number values should only be present

Number values should only be present

c. The list must be sorted
d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **13**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 90 and 99 ✓
- b. 89 and 94
- c. 89 and 99
- d. 90 and 94

Question **14**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(1)
- b. O👎
- c. O(log n) ✓
- d. O(n log n)

Question **15**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search can find multiple instances of the target element
- b. Binary search does not require dividing the list
- c. Binary search works on unsorted lists
- d. Binary search has a lower time complexity on large, sorted lists ✓